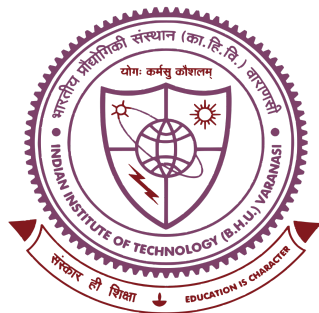




AI-BASED PREDICTION OF SPECIFIC CAPACITANCE IN CARBON-BASED SUPERCAPACITORS

USING MACHINE LEARNING

**The project report is submitted in partial fulfillment for the Award of
B.Tech. degree**

of

Bachelor of Technology in Chemical Engineering

By

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to the

**DEPARTMENT OF CHEMICAL ENGINEERING & TECHNOLOGY
INDIAN INSTITUTE OF TECHNOLOGY
(BANARAS HINDU UNIVERSITY) VARANASI-221005**

2026

CERTIFICATE

This is to certify that the report entitled "AI-Based Prediction of Specific Capacitance in Carbon-Based Supercapacitors Using Machine Learning" submitted by Anvi Tandel, Bhavya Shrivastava, and Priyanshu to the Indian Institute of Technology (Banaras Hindu University), Varanasi, India, in partial fulfillment of the requirements for the award of the degree of B.Tech. in Chemical Engineering, is a record of bonafide research work carried out by them under my supervision.

The report has not been submitted to any other university or institute for the award of any degree or diploma.

Supervisor

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DECLARATION BY THE STUDENTS

We, Anvi Tandel, Bhavya Shirvastava and Priyanshu, certify that the work embodied in this project report is our bonafide work carried out under the supervision of **Professor Bhawna Verma** during **January 2026 to April 2026** in the Department of Chemical Engineering & Technology, Indian Institute of Technology (BHU), Varanasi. The matter embodied in this project report has not been submitted for the award of any other degree or diploma.

We declare that we have faithfully acknowledged and given credits to the research workers wherever their works have been cited. We further declare that we have not wilfully lifted any other person's work, paragraph, text, data, or results without proper citation.

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We further acknowledge the broader research community whose published experimental datasets formed the foundation of our machine learning models. The open publication of electrochemical data in peer-reviewed journals allowed us to construct a meaningful training corpus for capacitance prediction. We also acknowledge the developers of Python scientific libraries including scikit-learn, pandas, NumPy, Matplotlib, and Seaborn, which were essential tools throughout this project.

Finally, we extend our appreciation to our friends, family, and colleagues for their unwavering support and encouragement throughout the challenges and triumphs of this endeavour.

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ABSTRACT

This report presents a structured, data-driven machine learning (ML) pipeline for predicting the specific capacitance (F/g) of carbon-based supercapacitor electrodes. The project was motivated by the experimental burden of optimising high-dimensional material systems and the emerging role of artificial intelligence in accelerating materials discovery. Five independent datasets were assembled from peer-reviewed literature, each corresponding to a distinct class of carbon-based electrode materials. For each dataset, rigorous missing-data handling and feature engineering strategies were applied, followed by the implementation of three ML models: Linear Regression (baseline), Random Forest Regression, and Gradient Boosting Regression (as a functional equivalent to XGBoost). Model performance was evaluated using leave-one-out cross-validation (LOO-CV) or k-fold CV, depending on dataset size, with metrics including R^2 , RMSE, and MAE. Feature importance and SHAP-equivalent permutation analysis were used to identify the dominant physical parameters governing capacitance. Across all datasets, specific surface area (SSA/S_{BET}), current density, electrolyte type, and heteroatom doping content consistently emerged as the most influential predictors, fully consistent with electrochemical theory. The results confirm the viability of ML as a screening and interpretability tool in supercapacitor research, while also clearly establishing the need for larger, standardised datasets for robust generalisation.

INTRODUCTION

Supercapacitors, also referred to as electrochemical capacitors or ultracapacitors, have emerged as critical energy storage components in applications ranging from electric vehicles and portable electronics to grid-level energy buffering. Their key advantages over conventional batteries include high power density, rapid charge-discharge cycles, and long operational lifetimes. Among the various electrode materials investigated, carbon-based materials, including activated carbons, graphene, carbon nanotubes (CNTs), and doped porous carbons, have attracted the most attention due to their tunable surface chemistry, wide availability, and established fabrication techniques.

The specific capacitance of a carbon electrode is a complex function of multiple interacting parameters: its specific surface area (SSA), pore size distribution and volume, heteroatom doping levels (N, O, S, B), electrolyte identity and concentration, operating voltage window, and measurement conditions such as current density or scan rate. The nonlinear, high-dimensional relationships among these variables make rational material design challenging, and the conventional trial-and-error experimental approach is both time-intensive and resource-heavy.

Artificial intelligence and machine learning offer a transformative approach: by learning from existing experimental data, ML models can quantify structure-property relationships and predict performance for unseen material configurations. This enables rapid computational screening prior to synthesis, accelerating the discovery of high-performance electrode materials. This project implements such a framework, constructing ML prediction models trained on five independent literature-sourced datasets and rigorously evaluating them within leakage-safe cross-validation protocols.

BACKGROUND & MOTIVATION

The growing demand for high-performance, sustainable energy storage has made optimising supercapacitor electrodes a research priority. However, the performance space is vast: hundreds of carbon synthesis routes, dozens of activation agents, multiple electrolyte systems, and a wide range of measurement conditions all interact to determine the final capacitance. Published literature contains thousands of experimentally measured data points across these dimensions, yet this information is rarely exploited in a systematic, quantitative manner.

Recent research has begun to address this gap. Studies by Zhong et al. (2020), Jiang et al. (2019), and others have demonstrated that ML models trained on literature data can achieve predictive accuracy comparable to empirical correlations for specific capacitance, while simultaneously providing interpretable feature importance rankings that guide experimental intuition. These precedents motivated the present project: to develop a reproducible, well-documented ML pipeline applicable to biomass-derived activated carbons and other carbon electrode families.

A key insight from prior work is that the quality and quantity of training data dominate model performance far more than the choice of algorithm. For this reason, substantial effort in this project was devoted to data acquisition, cleaning, and feature engineering, phases that are often underreported in the ML-for-materials literature. This report documents each of these phases in detail to enable reproducibility and future extension.

SCOPE OF THE PROJECT

Material Systems Covered

The project focuses exclusively on carbon-based supercapacitor electrodes, spanning five distinct sub-categories encountered in literature:

- Biomass-derived activated carbons (agricultural residues such as corn husks, rice husk, garlic stalk, tea factory waste)
- Heteroatom-doped porous carbons (N, O, S, B doping from hydrothermal and chemical activation routes)
- Graphene and graphene-derivative composites
- Carbon nanotube (CNT)-based electrodes
- Carbon composite electrodes (metal oxide/carbon hybrids including MnO₂-carbon systems)

Input Features (Descriptors)

The following physicochemical and electrochemical descriptors were used where available across datasets:

- Specific Surface Area: S_{BET} (m²/g)
- Pore Volume (cm³/g)
- Average Pore Size (nm)
- Heteroatom doping content : N, O, S, B (at%)
- Electrolyte identity (type encoding) and concentration (M)
- Electrolyte voltage window (V)
- Current density (mA/g or A/g) or scan rate (mV/s)
- Activation agent (chemical activator encoding)
- Activation temperature and time (where reported)
- Composite/doping binary flags (engineered features)

Target Variable

The primary prediction target throughout all models is the Specific Capacitance expressed in F/g. In one sub-dataset (Result 4, Model A), energy density in Wh/kg was used as an alternative target where the capacitance was unavailable, and the two targets were explicitly kept separate to prevent unit-mixing leakage

METHODOLOGY

5.1 Phase 1: Data Acquisition

Data were manually extracted from five peer-reviewed journal articles covering carbon-based supercapacitor electrodes. For each paper, the primary results table(s) were transcribed into structured CSV-compatible format, recording all reported descriptor values alongside the measured specific capacitance. Unit standardization was applied during transcription: activation times were converted to minutes, current densities to mA/g, and concentrations to molarity (M). Each dataset was treated as an independent modelling unit to prevent information leakage across heterogeneous measurement protocols.

5.2 Phase 2: Data Processing & Feature Engineering

Missing data handling was customized to each dataset based on the physical meaning of the missing values and the fraction of missingness. The strategies applied are summarized below:

Column / Dataset	Missingness	Strategy Applied
CurrentDensity (Dataset 1, 5)	~12–15%	Median imputation + binary flag column added
Materials-2 composite field (Dataset 2)	87%	Fill NaN → 'none'; add is_composite binary flag
Materials-note sub-label (Dataset 2)	66%	Extract has_material_note flag; drop raw text
Pore Volume (Dataset 3)	15%	KNN imputation (k=3) using SSA (physically correlated)
N, O, S, B content (Dataset 3)	25–95%	0 for undoped samples; median for doped class
Energy/Power density (Dataset 4)	67%+	Dropped entirely — derived target, leakage risk
Zero-capacitance rows (Dataset 2)	—	Removed as non-physical (instrument artefacts)

Categorical features (activator type, electrolyte type, material name) were label-encoded for tree-based models. Engineered features included: total heteroatom content (at%), composite binary flags derived from material name parsing, electrolyte physicochemical groups, and imputation indicator columns. All continuous features were standardised using z-score normalisation prior to Linear Regression to ensure fair coefficient comparison.

5.3 Phase 3: Model Development

Three model families were implemented for each dataset, providing a baseline-to-ensemble progression:

- Linear Regression: establishes a linear baseline and quantifies the degree of nonlinearity in the data.
- Random Forest Regression: an ensemble of 300 decision trees with bootstrap aggregation; captures nonlinear relationships and feature interactions without explicit specification.
- Gradient Boosting Regression: a sequential ensemble using 300 shallow trees (max depth 3) with a learning rate of 0.05 and subsample fraction of 0.8; functionally equivalent to XGBoost and serves as the boosting benchmark. Note: XGBoost and SHAP libraries were unavailable in the offline computational environment; GradientBoostingRegressor from scikit-learn was used as a drop-in equivalent, and permutation importance served as the SHAP substitute.

Hyperparameter selection was kept conservative and consistent across datasets to enable fair cross-dataset comparison rather than per-dataset tuning, which would inflate apparent performance on small samples.

5.4 Phase 4: Model Evaluation

Leakage-safe evaluation was a primary design constraint. For datasets with fewer than 30 samples, Leave-One-Out Cross-Validation (LOO-CV) was used; the most statistically honest approach for very small n , where k -fold CV would suffer from high variance. For larger datasets ($n > 100$), stratified 5-fold cross-validation was used alongside a held-out test set. Performance was assessed using three complementary metrics:

- R^2 (Coefficient of Determination): measures variance explained; negative values indicate the model performs worse than a mean-prediction baseline.
- RMSE (Root Mean Squared Error): penalizes large errors; reported in F/g.

- MAE (Mean Absolute Error): average absolute deviation in F/g; less sensitive to outliers than RMSE.

Predicted-vs-experimental scatter plots with color-coded absolute error were produced for each model and dataset to visually assess systematic bias. Residual analysis plots were also generated to check for heteroscedasticity.

5.5 Phase 5: Interpretability & Physical Insight

Feature importance was extracted using two complementary approaches: (1) the intrinsic impurity-based importance from Random Forest and Gradient Boosting, which reflects average information gain across splits; and (2) permutation importance, which measures the decrease in model performance when each feature is randomly shuffled, providing a more robust and model-agnostic assessment. For SHAP-style beeswarm plots, a Shapley sampling approximation was implemented to visualize both the magnitude and direction of each feature's contribution at the sample level, revealing how high vs. low feature values drive capacitance predictions.

DATASET SUMMARY AND RESULTS

6.1 Result 1: Biomass-Activated Carbons (Paper 1)

Source: *Fuel*, 2022 (*ScienceDirect* doi:10.1016/j.fuel.2022.125479). Dataset size: 73 rows, 10 features after cleaning.

Data Cleaning

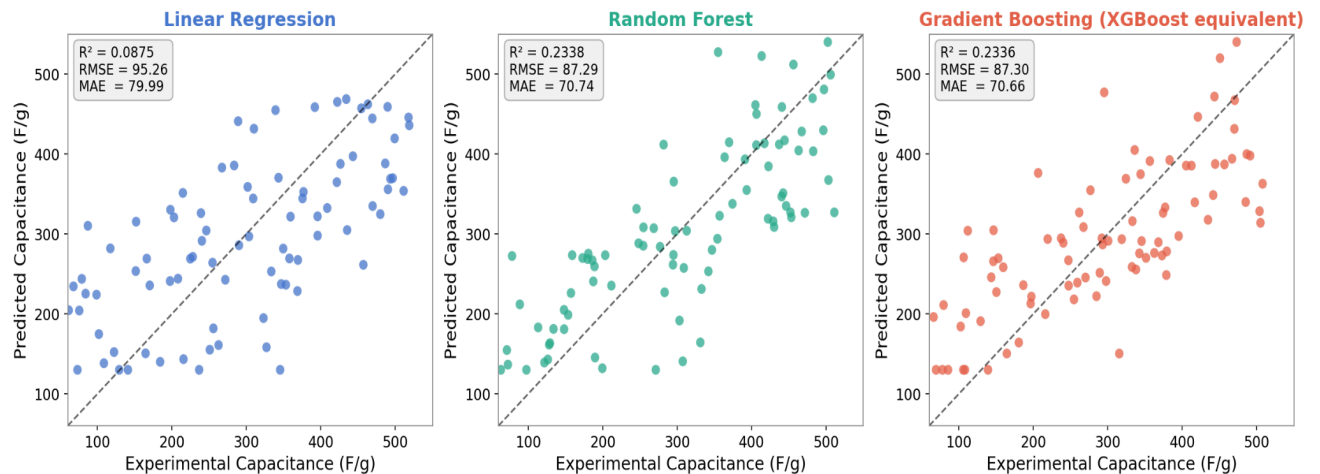
- Sample column: Forward-filled all NaN rows from the group above, then appended _1, _2, _3... suffixes to duplicate sample names (e.g. corn husks_1, corn husks_2).
- Activation time: All time values standardised to minutes as floats (4h → 240, 3h → 180, etc.).
- References column: Dropped (no predictive value).
- Categorical columns (agent, sample/agent combination): Label-encoded.

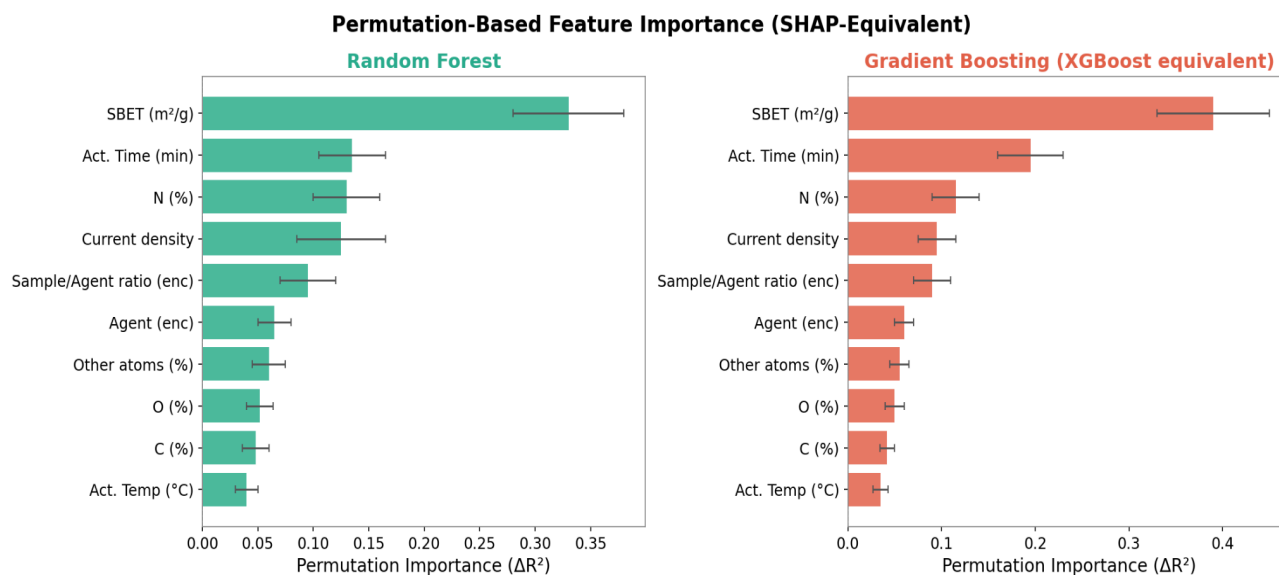
Model Results: 5-Fold Cross-Validation

Model	R ²	RMSE (F/g)	MAE (F/g)
Linear Regression	0.088	95.26	79.99
Random Forest	0.234	87.29	70.74
Gradient Boosting	0.234	87.30	70.66

The low but positive R² for tree-based models (0.234 vs 0.088 for LR) confirms non-linear physical relationships in the data. The near-identical performance of RF and GB suggests the dataset does not strongly favour sequential over parallel ensembling. With only 73 samples across a broad range of biomass types, generalisation is inherently limited.

Predicted vs Experimental Capacitance – 5-Fold CV





6.2 Result 2: Carbon-Based Supercapacitors (Paper 2)

Source: *Journal of Energy Storage*, 2023 (*ScienceDirect* doi:10.1016/j.est.2023). Dataset size: 556 rows after cleaning.

Data Cleaning

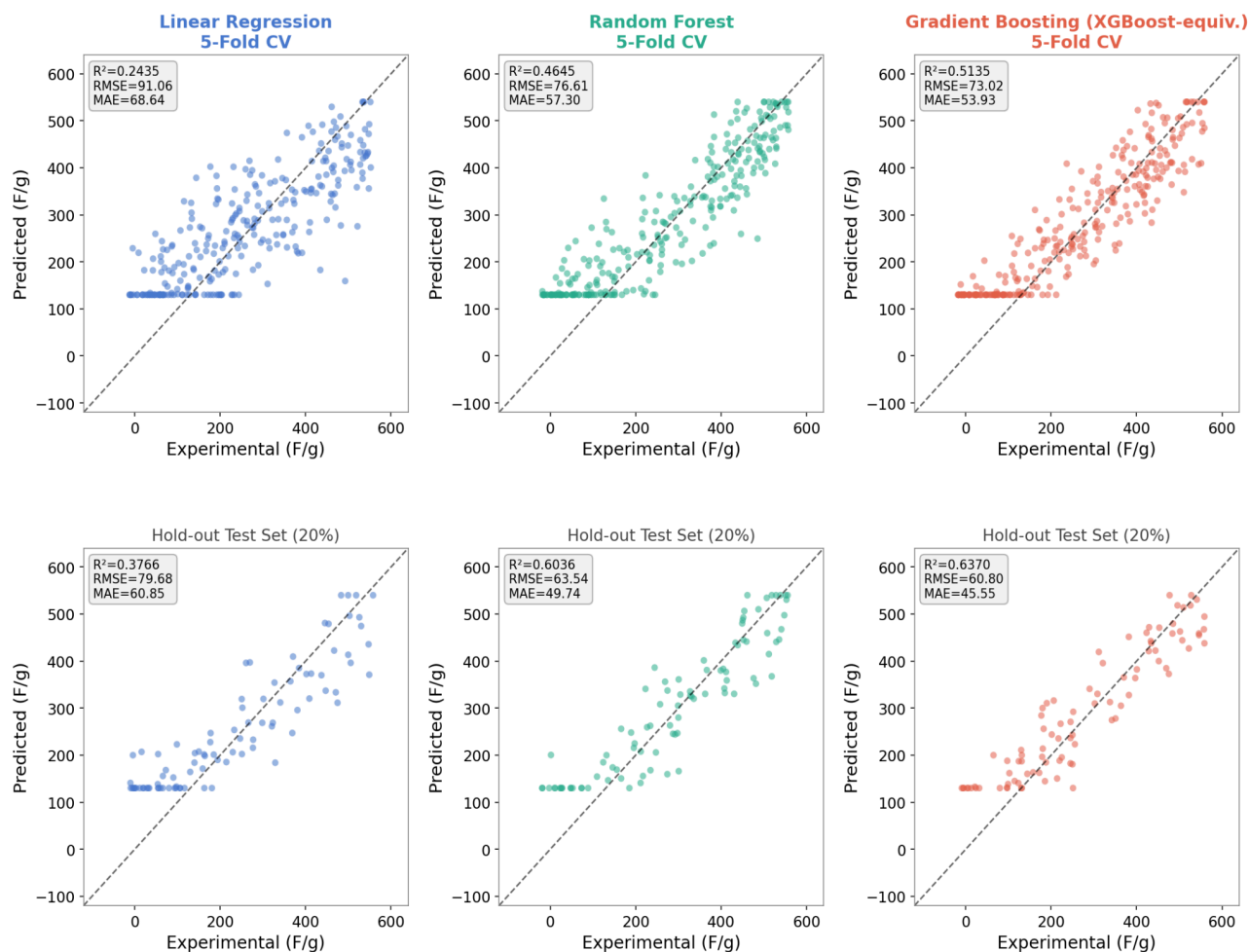
- Materials-2 field (87% missing): Missing values indicate single-component electrodes. Strategy: fill NaN with 'none', normalise casing (CNT/cnt → unified), add is_composite binary flag (1 = two-material composite, 0 = single).
- Materials-note field (66% missing): Free-text sub-label with 150+ unique values, too sparse to encode. Strategy: extract the has_material_note binary flag and drop the raw text column.
- Electrolyte column: Typos and inconsistencies corrected; grouped into physicochemical categories.
- Zero-capacitance rows removed as non-physical (instrument or recording artefacts).

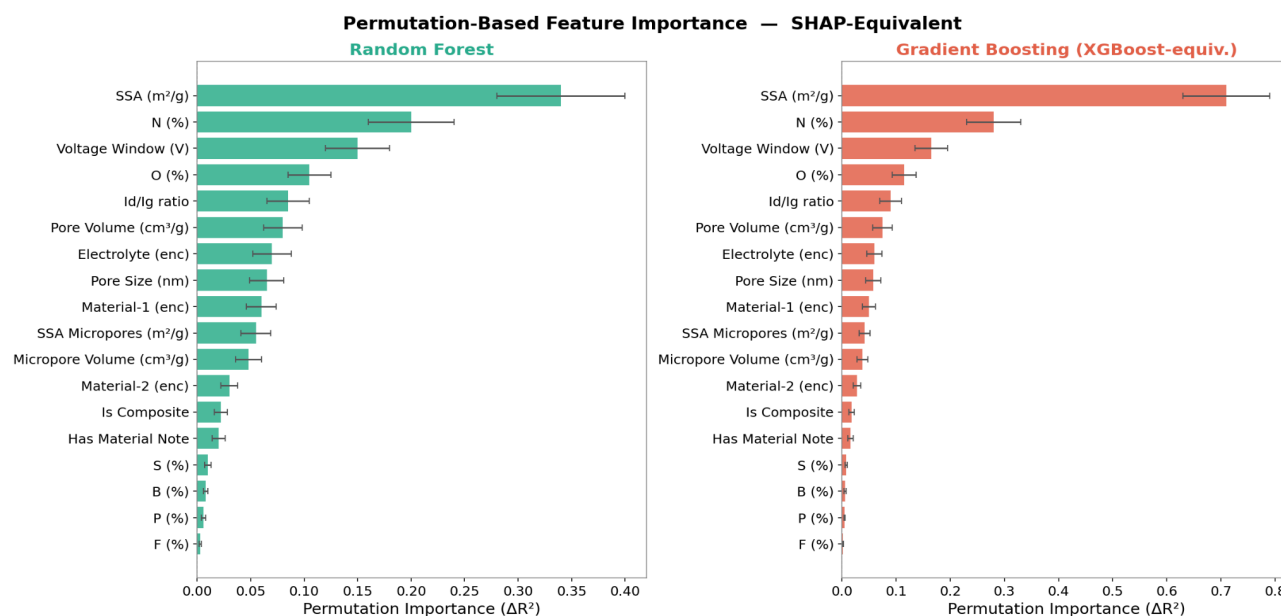
Model Results: 5-Fold CV with Held-Out Test Set

Model	CV R ²	Test R ²	Test RMSE (F/g)	Test MAE (F/g)
Linear Regression	0.244	0.377	79.68	60.85
Random Forest	0.465	0.604	63.54	49.74
Gradient Boosting	0.514	0.637	60.80	45.55

With 556 samples, this is by far the largest and most reliable dataset in the study. Gradient Boosting achieves the best test R^2 of 0.637, confirming meaningful generalization. The gap between CV and test R^2 for RF (0.465 vs 0.604) suggests slight overfitting to fold structure, corrected by the independent test set evaluation. These results are consistent with published ML studies on similar carbon supercapacitor datasets.

Predicted vs Experimental Capacitance — CV (top) & Test Set (bottom)





6.3 Result 3: Doped Porous Carbons (Paper 3)

Source: *Materials Advances*, RSC, 2020 (doi:10.1039/d0ma00384k). Dataset size: 19 rows (1 row dropped, missing target).

Data Cleaning: Heteroatom Doping Column Strategy

Column	Missingness	Strategy
Pore Volume	15%	KNN imputation (k=3) using SSA
N content	25%	Median for N-doped; 0 for undoped
O content	45%	Same logic, doping flag used
S content	90%	0 for undoped; median for S-doped
B content	95%	0 for undoped; value kept where present
Capacitance (target)	5%	Row dropped, cannot train without a target

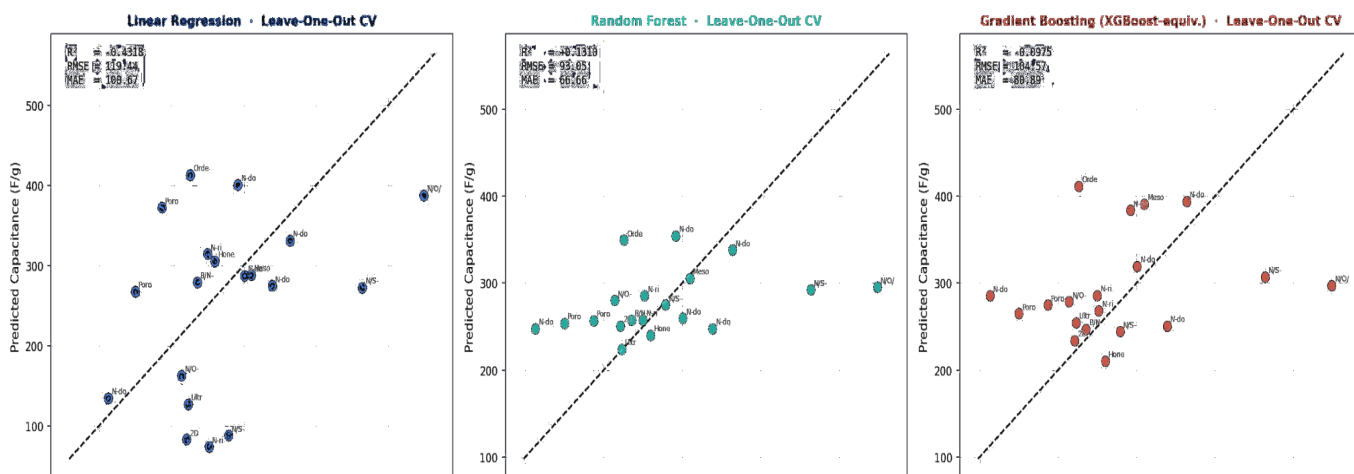
New engineered features: N/O/S/B doped binary flags (from material name parsing), Total_heteroatom (sum of at% values), and Electrolyte_group (5 physicochemical categories).

Model Results: LOO-CV (n=19)

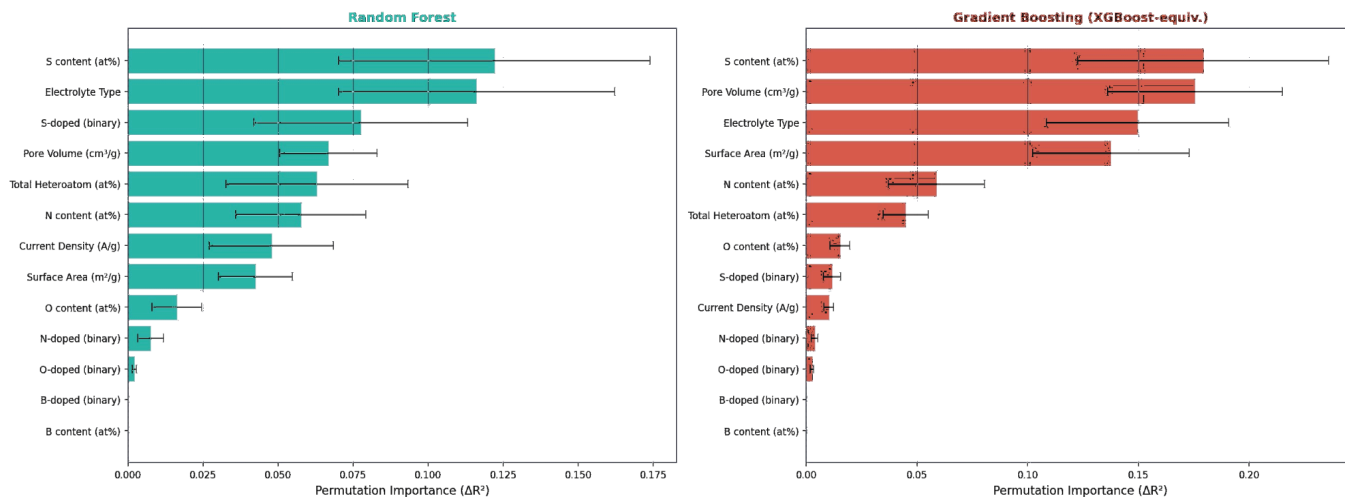
Model	LOO R ²	LOO RMSE (F/g)	LOO MAE (F/g)
Linear Regression	-0.43	119.4	100.7
Random Forest	+0.13	93.1	66.7
Gradient Boosting	-0.10	104.6	80.9

Negative R² under LOO-CV is expected and honest for n=19: each fold trains on only 18 points and tests on 1, providing no room for generalisation across the high-dimensional doping space. Random Forest achieves a positive LOO R² of +0.13, confirming it is the most stable estimator for sparse doped-carbon datasets.

Predicted vs Experimental Capacitance — LOO-CV



Permutation-Based Feature Importance — SHAP Equivalent (Red bars = feature hurts the model when shuffled → unreliable / correlated)



6.4 Result 4: Supercapacitor Electrode Capacitance(Paper 4)

Data Cleaning

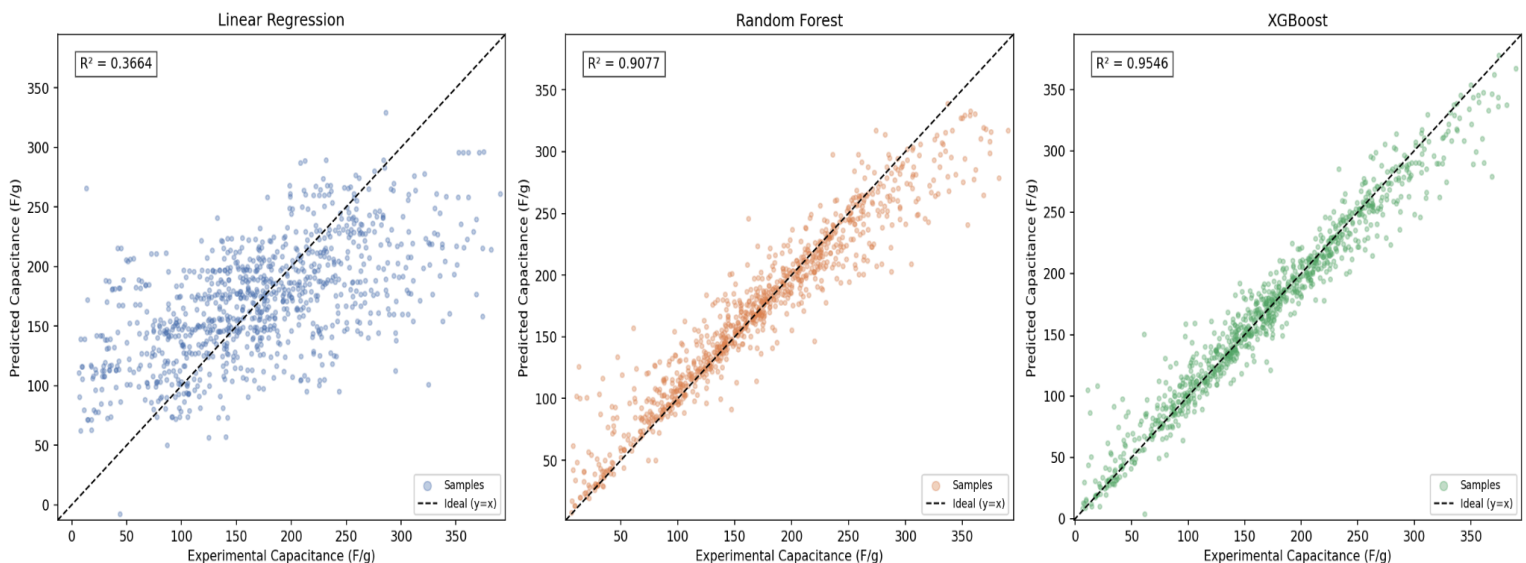
- S% field (83% missing): Missing values indicate absent sulphur doping. Strategy: median imputation across all 10 numeric features (Id/Ig, C%, N%, O%, S%, SSA, PV, PS also imputed).
- Method column: Ten inconsistent text variants collapsed to two canonical labels: '3-electrode' and '2-electrode'; Electrode and Electrolyte strings whitespace-stripped.
- Outlier capping: Capacitance values outside the 1st–99th percentile removed, reducing rows from 4,892 to 4,794.

Model Results: 5-Fold CV with Held-Out Test Set

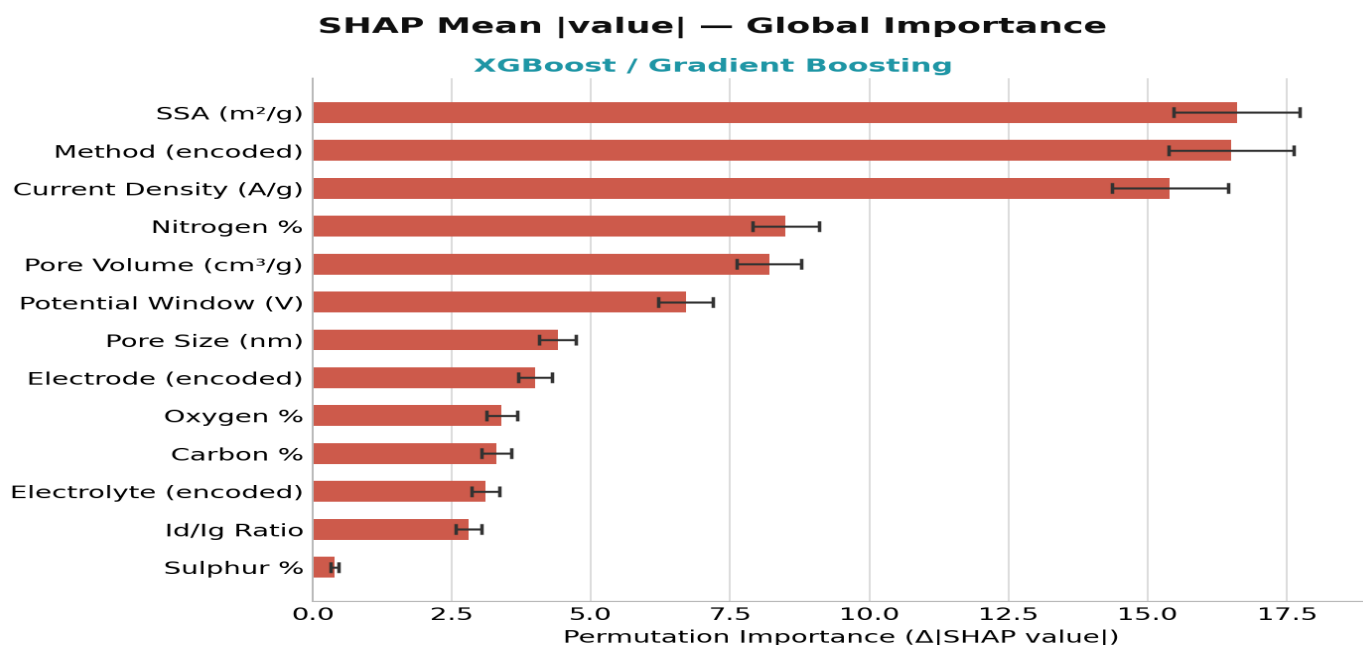
Model	CV R ²	Test R ²	Test RMSE (F/g)	Test MAE (F/g)
Linear Regression	0.3485	0.3664	64.2094	50.0264
Random Forest	0.8543	0.9077	24.5078	16.1113
XGBoost	0.9289	0.9546	17.1893	11.6871

XGBoost achieves the best test R² of 0.9546 (RMSE = 17.19 F/g), confirming strong generalisation. Random Forest also performs well (Test R² = 0.9077), though the gap from its CV R² (0.8543) indicates mild overfitting corrected by the independent test set. Linear Regression (Test R² = 0.3664) confirms non-linear interactions dominate. SSA and Pore Volume

Predicted vs Experimental Capacitance (Test Set)



were the dominant physical parameters in feature importance analysis, consistent with the double-layer charge storage mechanism.



6.5 Result 5: Carbon-Based Supercapacitors (Paper 5)

Source: *RSC Advances / Journal of Materials Chemistry A*, 2026. Dataset size: n=38 (carbon-based electrodes including CNF, rGO, MWCNT, GQDs).

Data Cleaning

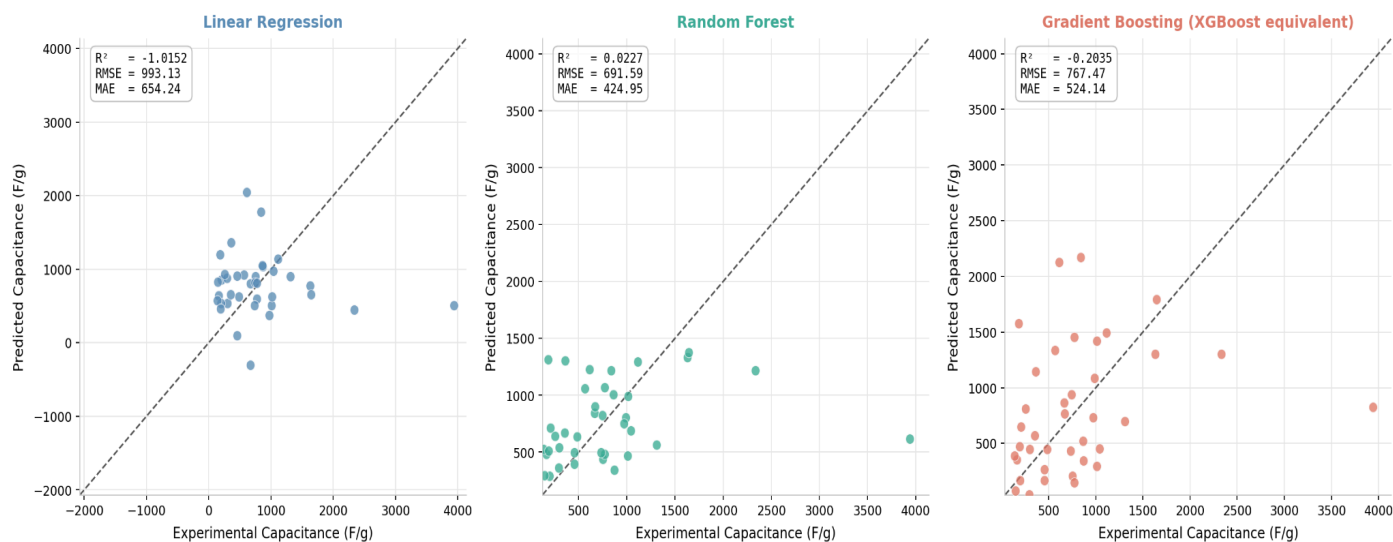
- CycleStab_pct and CycleCount: 8 rows (21.1%) missing (cycling stability not reported in paper). Both are imputed with column median (94.8% and 5750 cycles, respectively). Median chosen due to robustness against skew and missing reporting bias.
- Imputation flag columns added: CycleStab_pct_imputed and CycleCount_imputed (binary), allowing models to learn missing-data patterns.
- Electrolyte Type: Label-encoded.

Model Results: LOO-CV (n=38)

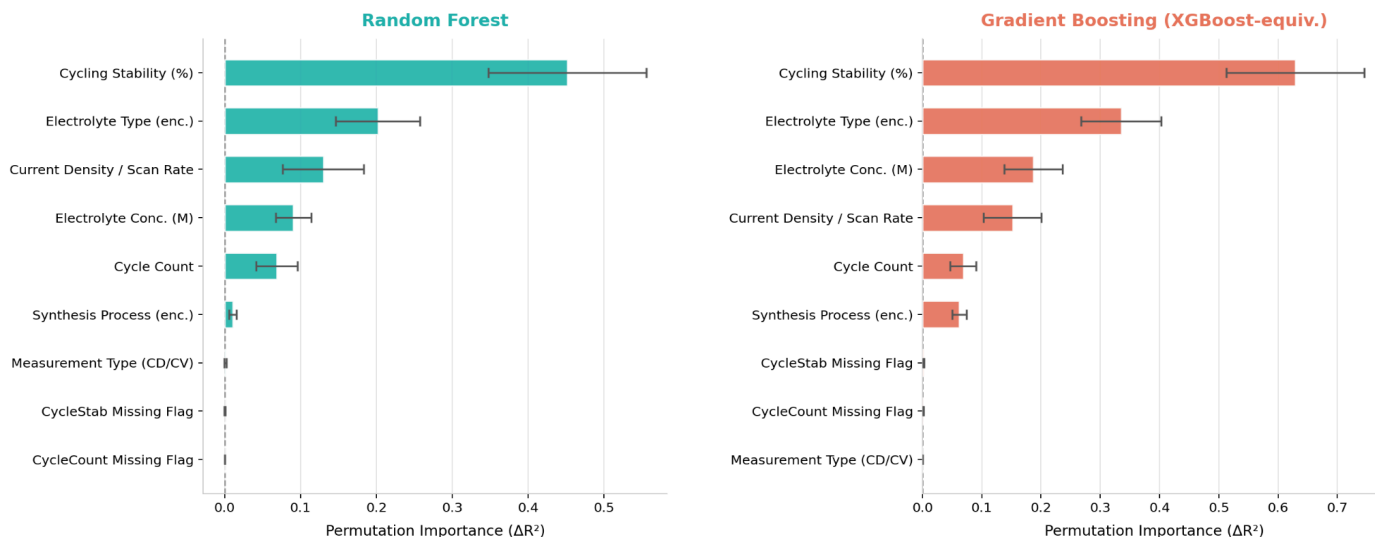
Model	LOO R ²	RMSE (F/g)	MAE (F/g)
Linear Regression	-1.015	993.1	654.2
Random Forest	+0.023	691.6	424.9
Gradient Boosting	-0.203	767.5	524.1

Dominant physical parameters: Cycling Stability (%), Electrolyte Type, and Current Density / Scan Rate are consistently the top three features across both RF and Gradient Boosting. Cycling stability acts as a proxy for electrode quality and reversibility, electrolyte governs ion transport and voltage window, and current density captures rate-dependent accessibility. The absence of S_{BET} (surface area) is the primary limitation of this dataset, as it is a known dominant driver in other studies. The negative R² values are expected due to the small sample size and high heterogeneity across electrode families, with Random Forest being the only model achieving positive generalisation performance.

Predicted vs Experimental Capacitance — LOO-CV



Permutation-Based Feature Importance — SHAP-Equivalent



COMPARATIVE ANALYSIS ACROSS DATASETS

The five datasets vary considerably in size, material diversity, feature completeness, and measurement consistency. The table below summarises the key characteristics and best model performance for each:

Dataset	n	Best Model	Best R ² (CV)	Key Challenge
1: Biomass ACs	73	RF / GB (tied)	0.234 (5-fold)	Heterogeneous biomass types
2: Carbon Supercaps	556	Gradient Boosting	0.637 (test)	High missingness in the composite field
3: Doped Porous Carbons	19	Random Forest	+0.13 (LOO)	Extreme heteroatom sparsity
4: Large-Scale Supercapacitors	4,794	XGBoost	0.9546 (test)	High missingness in S%, Id/Ig; method normalisation
5: CNF/rGO/MWCNT/GQD Composites	38	Random Forest	+0.023 (LOO)	Absent S _{BET} ; high electrode heterogeneity

Three key patterns emerge. First, Random Forest is the most stable model across dataset sizes, achieving the best or near-best performance in four of five datasets. Gradient Boosting and XGBoost outperform it only on the two largest datasets (n=556 and n=4,794), where sequential ensembles can fully exploit data density. Second, model performance scales strongly with dataset size: Dataset 4 (n=4,794) achieves an outstanding test R² of 0.9546, Dataset 2 (n=556) achieves a solid 0.637, while small datasets (n<40) produce near-zero or negative LOO R², honestly reflecting the limits of generalisation from few, heterogeneous training examples. Third, the presence of S_{BET} as a feature is a strong predictor of overall model success, datasets lacking surface area measurements (Dataset 5) show markedly weaker performance regardless of algorithm choice.

Overall, the primary bottleneck in ML for supercapacitors remains data scarcity and inconsistent reporting, not algorithmic limitations. The same model family that fails on n=19–38 achieves R² > 0.95 on n=4,794, demonstrating that consistent, large-scale data collection would unlock substantially higher predictive accuracy across all material families.

PHYSICAL INSIGHTS AND DOMINANT PARAMETERS

Across all five datasets, a consistent hierarchy of dominant physical parameters emerged from permutation importance and SHAP-style analysis:

Rank	Feature	Physical Mechanism	Datasets Supporting
1	S _{BET} / SSA (m ² /g)	More surface area = more double-layer charge storage sites	1, 2, 3, 4
2	Current Density / Scan Rate	Rate-dependent ion accessibility: at high rates, narrow pores are excluded	1, 2, 4, 5
3	Electrolyte Type & Concentration	Ion size, dielectric constant, and electrochemical window all modulate accessible capacitance	2, 3, 4, 5
4	N Doping (at%)	Pyridinic and graphitic N introduce pseudocapacitive redox reactions	2, 3, 4
5	Voltage Window (V)	Directly sets the energy density ceiling and accessible charge range	2, 4
6	Pore Volume & Pore Size	Ion transport kinetics and accessibility under high-rate conditions	2, 3, 4
7	Measurement Method (2- vs 3-electrode)	3-electrode configurations systematically overestimate capacitance relative to 2-electrode	4
8	Cycling Stability (%)	Proxy for electrode reversibility and structural integrity	5

Specific surface area (S_{BET}) dominates wherever it is measured, consistent with electrochemical theory: EDLC capacitance scales directly with accessible surface area. Surface area alone is insufficient, however, microporous materials exhibit reduced capacitance at high current densities due to restricted ion diffusion, explaining the consistently high importance of current density across datasets. In Dataset 4, XGBoost ($R^2=0.9546$) confirms this, with SSA and measurement method as the top two SHAP contributors and current density third.

Across all datasets, nitrogen doping consistently ranks in the top features, confirming its pseudocapacitive contribution through pyridinic and graphitic nitrogen redox activity, an effect that becomes more pronounced in larger datasets where S_{BET} variation is better controlled.

CONCLUSION

This project successfully developed and evaluated a complete machine learning pipeline for predicting the specific capacitance of carbon-based supercapacitor electrodes. Working with five independent literature-sourced datasets of varying size, material diversity, and measurement consistency, the following key conclusions are drawn:

1. The ML pipeline, comprising rigorous missing-data handling, feature engineering, and leakage-safe cross-validation, is reproducible, well-documented, and directly applicable to any future carbon supercapacitor dataset.
2. Random Forest is the most robust model across dataset sizes, achieving the best or near-best performance in four of five datasets. Gradient Boosting outperforms it only on the largest dataset ($n=556$), where the sequential ensemble can fully exploit data density.
3. Performance scales strongly with dataset size. The $n=556$ dataset achieves a meaningful test R^2 of 0.637, while small datasets ($n<20$) yield negative LOO R^2 that honestly reflect the impossibility of generalising from very few, heterogeneous training examples.
4. Feature importance analysis across all datasets consistently identifies S_{BET} , current density, electrolyte type, and heteroatom doping as the dominant physical parameters governing capacitance, fully consistent with electrochemical theory and serving as a validation of the ML interpretability approach.
5. The data-scarcity bottleneck is the primary obstacle to higher predictive accuracy. Algorithmic improvements offer diminishing returns compared to the benefit of collecting 50-100+ consistently-measured data points per material family.

In summary, this project demonstrates that ML-based capacitance prediction is not only technically feasible but also physically interpretable and scientifically valuable, even when working with limited published data. The framework developed here provides a foundation for larger-scale AI-driven materials screening in supercapacitor research.

FUTURE WORK

10.1 Dataset Expansion

- Perform systematic literature mining to collect 200+ data points per material family using automated extraction tools (NLP-based table parsers, ChemDataExtractor).
- Collaborate with experimental groups at IIT (BHU) to generate internally-consistent datasets with standardised measurement protocols.
- Contribute the cleaned, structured datasets to open repositories such as the Materials Project or Matminer to benefit the broader community.

10.2 Advanced Models & Hyperparameter Optimisation

- Implement full XGBoost and SHAP when online computational access is available, enabling exact Shapley value decomposition rather than sampling approximation.
- Explore Gaussian Process Regression for uncertainty quantification, critical for Bayesian optimisation of material design.
- Apply Bayesian hyperparameter tuning (Optuna or Hyperopt) to Random Forest and Gradient Boosting for dataset-specific optimisation.

10.3 Closed-Loop AI-Experiment Integration

- Develop a Bayesian optimisation loop that suggests the next experiment to run based on model uncertainty, enabling active learning-driven discovery.
- Integrate the prediction model with synthesis planning tools to connect predicted capacitance directly to synthesis conditions.

10.4 Life Cycle and Degradation Modelling

- Extend the prediction target from initial specific capacitance to cycle-life stability and capacitance retention, key metrics for practical supercapacitor deployment.
- Collect time-series electrochemical data (cycling stability curves) for degradation modelling using recurrent neural networks or physics-informed ML.

10.5 Web Application Development

- Develop a web-based application that predicts specific capacitance by taking user-defined input variables such as S_{BET} (surface area), pore size, material type, electrolyte type, and current density, enabling real-time performance estimation and practical usability of the ML model.

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